Module 1

Review of Securities and Security Markets

August 27, 2025

Securities

- Personal finance:
  - loan for a new car from a bank (secured)
    - principal, interest
    - pay back schedule
    - collateral (the car)
  - loan for a luxury cruise from your uncle (unsecured)
- Corporate finance:
  - bank loans – usually expensive
  - bonds – loan from investors
    - principal, coupons (interest), maturity
    - "fixed income" instruments
    - can be either secured or unsecured
  - stocks – shares of ownership in a company (equity)
    - voting right
    - dividend – share of profit

Securities: "legal right for future benefits"

"Legal right":

- promise
- priority – example: senior bonds, junior bonds, preferred stocks, stocks

Transfer of securities → security markets

- Primary markets: companies → investors (e.g., IPO)
- Secondary markets: investors ←→ investors

  e.g., NYSE, NASDAQ, etc
- "Market makers" – liquidity providers
  - bid/ask (offer)
    - bid – max price a dealer willing to buy
    - ask (offer) – min price a dealer willing to sell

    implicit size restrictions
  - positions
    - long
    - short
- Significance of bid-ask spread:
  - liquidity
  - trading cost
- Shorting:
  - asset borrowing, borrow cost
  - cash flows from the asset (e.g., dividend) – return to owner
    - margin requirement
- Market indices: measure of overall market performance

  Major US equity market indices:
  - Dow Jones Industrial Average (DJIA)
  - S&P 500 Index (SPX)
  - NASDAQ Composite Index (COMP)
  - Russell 2000 Index (RUT)
  - …

Effect of dividend payment (D) on stock price (S):

Assume a cash dividend D paid to the share S, since the share holders' net wealth does not change right before and after the dividend payment,

$S \rightarrow S - D$

immediately after dividend payment

Derivatives

A financial contract is a derivative security, or a contingent claim, if its value at expiration date T is determined exactly by the market price of the underlying cash instrument at time T (Ingersoll).

Examples:

- forwards – agreements to buy/sell underlying asset at pre-specified price and date
- futures – exchange traded forwards
- options – right but not obligation to buy/sell underlying asset (S) at pre-specified price (K) and date(s)
    - call option payoff: max(S – K, 0)
    - put option payoff: max(K – S, 0)
- others …

These are examples of products of financial engineering, which are built using primary assets such as equity, bonds, cash, commodity, …

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The first ever option trade: 600 BC by Greek philosopher Thales of Miletus

"…from his knowledge of astronomy he had observed while it was still winter that there was going to be a large crop of olives, so he raised a small sum of money and paid round deposits for the whole of the olive-presses in Miletus and Chios, which he hired at a low rent as nobody was running him up; and when the season arrived, there was a sudden demand for a number of presses at the same time, and by letting them out on what terms he liked he realized a large sum of money, so proving that it is easy for philosophers to be rich if they choose"

(quote from "Politics" by Aristotle)

Option markets in post pandemic world:

"Eighty Percent of the World's Stock Options Aren't Traded Where You Think": WSJ, March 11, 2024 (https://www.wsj.com/finance/stocks/eighty-percent-of-the-worlds-stock-options-arent-traded-where-you-think-35a46ddc?mod=saved_content)

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Examples of usage of derivatives (more detail later in the course):

- an airline co uses crude oil forwards/futures to "hedge" fuel price risk
- a speculator uses call options to augment his bets on stock returns
- lock in stock price using zero cost collar

In this course we will mainly concerned with "*financial derivatives*". Furthermore, to illustrate the general principle we will mostly confine ourselves within the "*equity derivatives*".

Financial Engineering

- hedging, custom solutions to complex risk management problems
- arbitrage
- speculation – leveraged bet
- tax, accounting, regulation
- effects on the cash markets
    - indication for market sentiment
    - feedback effects (more later in the course)

Financial Engineers

- Sell side: manufacturers
    - right products at right price for clients
    - risk management

Careers:

- Model building/development ("classic quants")
- System implementation/optimization ("analytics")
- Structuring
- Sales & trading
- Risk management
  - risk modeling
  - risk monitoring & control
- Control & regulation
  - model review
  - model governance
  - auditing

ML are playing increasingly important roles …

- Buy side: sophisticated users
  - identify the most appropriate products at the most attractive price for their needs

    → may propose product specs to sell side
  - risk management

Beyond Finance

- Optionality is everywhere ….

  → recognize its value
- Real options